

The empirical recurrence for OEIS A252771, recovered from its tail

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Abstract

OEIS A252771 records “Empirical recurrence of order 63” and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 135$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 2$, so the recurrence is proved for every $n \geq 65$. Its last coefficient is $c_{63} = -61917364224 \neq 0$, so no further tail satisfies anything shorter; any order-63 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry’s recurrence is the one displayed here, whatever its linked file contains, and it is proved.

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1 A conjecture that is not written down

OEIS A252771 is “Number of $(n+2) \times (1+2) \dots 3$ arrays with every consecutive three elements in every row, column and nw-se diagonal having exactly two distinct values, and new values 0 upwards introduced in row major order.”. It has offset 1 and begins

234, 1028, 4687, 22295, 105956, 507049, 2429173, 11652188, ...

The entry states:

Empirical recurrence of order 63 (see link above)

As of the “Last modified” line on the live entry (Jul 23 2025 13:38:34, revision 7) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry’s condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u , and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) = \iota^\top M^{j(n)} \tau$$

for a linear reindexing j fixed by the entry’s shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 135$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 135$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S' + 1$ consecutive values of a . $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 63: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 2$ is the first whose minimal order is exactly the stated 63.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 270$ exact terms from $n = 2$ on, it returns order 63 — the order the entry states — with integer coefficients:

$$a(n) = \sum_{i=1}^{63} c_i a(n-i),$$

c_1	5	c_{17}	21615184	c_{33}	−3230677492	c_{49}	−9916848576
c_2	13	c_{18}	44593296	c_{34}	3675232328	c_{50}	−647475277536
c_3	−8	c_{19}	−585874	c_{35}	7966451128	c_{51}	−213933640896
c_4	−339	c_{20}	−112940393	c_{36}	−29709694350	c_{52}	84261259008
c_5	−543	c_{21}	−110248943	c_{37}	−87730647932	c_{53}	−372543446016
c_6	2557	c_{22}	119364080	c_{38}	−4393616194	c_{54}	−203859666432
c_7	9640	c_{23}	142500675	c_{39}	146722387612	c_{55}	551049661440
c_8	2299	c_{24}	−278041573	c_{40}	123137176972	c_{56}	915547650048
c_9	−68751	c_{25}	150090912	c_{41}	−16609210524	c_{57}	30310723584
c_{10}	−153922	c_{26}	1585428329	c_{42}	−106050480164	c_{58}	−1041507237888
c_{11}	153773	c_{27}	996985931	c_{43}	−27478516352	c_{59}	−396777553920
c_{12}	915896	c_{28}	−3702474838	c_{44}	−89830642360	c_{60}	534538911744
c_{13}	1285383	c_{29}	−7351459707	c_{45}	−359542515960	c_{61}	261858852864
c_{14}	−2335021	c_{30}	1828926090	c_{46}	−73325316816	c_{62}	−113515167744
c_{15}	−9087078	c_{31}	12937285641	c_{47}	544993869024	c_{63}	−61917364224
c_{16}	−3909322	c_{32}	4609007481	c_{48}	644418484752		

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 65$, and it is the recurrence OEIS A252771 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 2$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{63} - c_1 x^{62} - \dots - c_{63}$. Its constant term is $-c_{63} = 61917364224$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k \lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component

of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 63 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 63, so $\deg q = 63$, and it is monic. It holds from some index t on. From $\max(t, 2)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 63, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 19 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 63, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry’s.

Third, the recovered recurrence was evaluated directly on the entry’s own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is -61917364224 .

Remark 4. The conjecture’s own statement was never read. The entry pins it down to the family of order-63 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

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